

Zhicheng (Stark) Guo

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SUMMARY

Machine learning researcher developing foundation models and interpretable AI for multimodal health data, including wearable physiological signals and medical imaging. My work spans representation learning, multimodal modeling, large-scale evaluation, and human-interpretable AI for health applications, with research published in NEJM AI, Nature Machine Intelligence, JMLR, CVPR, WACV, and HDSR.

EDUCATION

Duke University

Durham, NC

Ph.D. in Electrical and Computer Engineering (Advisor: Cynthia Rudin) · Jan 2022 – May 2026

M.S. in Computer Science (transferred to Ph.D. program) · Aug 2020 – Dec 2021

Rensselaer Polytechnic Institute

Troy, NY

B.Sc. in Computer Science · Jul 2016 – May 2020

EXPERIENCE

Postdoctoral Researcher — Duke University

Jul 2026 – Present

- Developing foundation models for large-scale medical imaging, including breast (mammography) and lung cancer risk prediction.
- Developing methods to extract interpretable concepts from foundation-model representations and evaluating their predictive and scientific utility.
- Building large-scale data-processing, model-training, and evaluation pipelines for reproducible medical-AI research.

Machine Learning Research Intern — Texas Instruments, Kilby Labs

May 2025 – Aug 2025

- Architected and deployed a self-evolving LLM-agentic optimization framework for hardware-constrained neural architecture search and mixed-precision quantization on edge hardware.
- Implemented multi-objective reward functions and on-the-fly hardware profiling to guide optimization decisions and improve real-time performance.
- Developed specialized small reasoning language model groups (SLMs) to replace generic LLM agents using GRPO and fine-tuning, enabling on-site, data-secure inference at a fraction of the compute cost.

Quantitative Research Intern — Duke University Health System

May 2021 – Aug 2021

- Modeled correlations between CSF/PBMC cellular populations and markers and cognitive scores in HIV-associated neurocognitive disorder patients across drug-use conditions.
- Built machine learning models on flow-cytometry data to predict drug use and impairment, supporting clinical assessment of drug effects.

PUBLICATIONS

(* denotes equal contribution / co-first authorship)

Zhicheng Guo, Cheng Ding, Duc H. Do, Amit Shah, Randall J. Lee, Xiao Hu, Cynthia Rudin. “SiamAF: Improving Atrial Fibrillation Detection Using a Shared Latent Space for ECG and PPG Signals.” Harvard Data Science Review (HDSR), 2025.

Alina Jade Barnett*, **Zhicheng Guo***, Jin Jing*, Wendong Ge, Brandon Westover, Cynthia Rudin. “Improving Clinician Performance in Classification of EEG Patterns on the Ictal-Interictal-Injury Continuum Using Interpretable Machine Learning.” NEJM AI, 2024.

Sully F. Chen, **Zhicheng Guo**, Cheng Ding, Xiao Hu, Cynthia Rudin. “Learned Kernels for Sparse, Interpretable, and Efficient Medical Time Series Processing.” Nature Machine Intelligence, 2024.

Cheng Ding, **Zhicheng Guo**, Cynthia Rudin, Ran Xiao, Amit Shah, Duc H. Do, Randall J. Lee, Gari Clifford, Fadi B. Nahab, Xiao Hu. “Learning From Alarms: A Robust Learning Approach for Accurate PPG-Based Atrial Fibrillation Detection Using Eight Million Samples Labeled with Imprecise Arrhythmia Alarms.” IEEE Journal of Biomedical and Health Informatics (JBHI), 2024.

Cheng Ding, **Zhicheng Guo**, Zhaoliang Chen, Randall J. Lee, Cynthia Rudin, Xiao Hu. “SiamQuality: A ConvNet-Based Foundation Model for Photoplethysmography Signals.” Physiological Measurement, 2024.

Zhicheng Guo, Cheng Ding, Xiao Hu, Cynthia Rudin. “A Supervised Machine Learning Semantic Segmentation Approach for Detecting Artifacts in Plethysmography Signals from Wearables.” Physiological Measurement, 2021.

Anish Karpurapu, **Zhicheng Guo**, Xiao Hu, Cynthia Rudin. “Data-Hugging Shields Proprietary AI Models from Research That Could Disprove Them.” npj Artificial Intelligence 2, 45 (2026).

Zhicheng Guo, Maximillian Machado, Jon Donnelly, Cynthia Rudin. “ProtoCDisco: Unsupervised Concept Discovery Using Pretrained Representations.” Submitted, NeurIPS 2026.

Varun Babbar*, **Zhicheng Guo***, Cynthia Rudin. “*What Is Different Between These Datasets? A Framework for Explaining Data Distribution Shifts.*” Journal of Machine Learning Research (JMLR), 2025.

Jon Donnelly, **Zhicheng Guo**, Alina Jade Barnett, Hayden McTavish, Chaofan Chen, Cynthia Rudin. “*Rashomon Sets for Prototypical-Part Networks: Editing Interpretable Models in Real-Time.*” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.

Luke Moffett, Frank Willard, Maximillian Machado, Emmanuel Mokel, Jon Donnelly, **Zhicheng Guo**, Adam Costarino, Julia Yang, Giyoung Kim, Alina Jade Barnett, Cynthia Rudin. “*Cosine Distance Is Almost All You Need (for Prototypical-Part Models).*” IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026.

Ruixin Zhang, Jon Donnelly, **Zhicheng Guo**, Ghazal Khalighinejad, Haiyang Huang, Alina Jade Barnett, Cynthia Rudin. “*It’s LIT! LLMs with Interpretable Tools.*” MTI-LLM Workshop, NeurIPS, 2025.

Manickam Ashokkumar, Wenwen Mei, Jackson J. Peterson, Yuriko Harigaya, David M. Murdoch, David M. Margolis, Caleb Kornfein, Alex Oesterling, **Zhicheng Guo**, Cynthia D. Rudin, Yuchao Jiang, Edward P. Browne. “*Integrated Single-Cell Multiomic Analysis of HIV Latency Reversal Reveals Novel Regulators of Viral Reactivation.*” Genomics, Proteomics & Bioinformatics, 2024.

TECHNICAL SKILLS

Languages: Python

Frameworks & training: PyTorch; GPU and distributed training on accelerators

Models: foundation models; Vision Transformers; ConvNets; prototypical-part networks; LLMs / small language models (SLMs)

Methods: representation learning; multimodal learning; interpretable-by-design ML; model evaluation and controlled ablations; reinforcement learning (GRPO); fine-tuning; neural architecture search; quantization

Domains: physiological signals (ECG, PPG, EEG); wearable sensor data; medical imaging